Swastik Majumder

Curriculum Vitae

Department of Physics &
Center for Cosmology and AstroParticle Physics (CCAPP)
The Ohio State University, Columbus, Ohio, United States

Education

Doctor of Philosophy in Physics 2025–Present
The Ohio State University, Columbus, Ohio, United States.

Bachelor of Science and Master of Science in Physics 2020–2025
Indian Institute of Science Education and Research Kolkata, India.

Research Interests

In-medium quantum field theory: Perturbative quantum chromodynamics at finite temperature and density through effective field theories.

Statistical mechanics: Equilibrium and nonequilibrium behavior of classical many body systems.

Publications

5. **S. Majumder** and T. Gorda.
Hot and dense QCD at N^3LO: infrared sector.
In preparation (2026).

4. **S. Majumder** and T. Gorda.
Hot and dense QED at N^3LO: infrared sector.
Preprint (2026).
arXiv:2610.10771.

3. **S. Majumder** and M. Barma.
Stochastic systems with Bose-Hubbard interactions: Effects of bias on particles on a random comb.
Physical Review E **113**, 044137 (2026).
Journal | arXiv:2601.09962.

2. **S. Majumder** and M. Barma.
Stochastic systems with Bose-Hubbard interactions: Effects of bias on particles on a one-dimensional lattice.
Physical Review E **113**, 044136 (2026).
Journal | arXiv:2508.03344.

1. **S. Majumder** and N. Sasakura.
Three cases of complex eigenvalue/vector distributions of symmetric order-three random tensors.
Progress of Theoretical and Experimental Physics **2024**, 093A01 (2024).
Journal | arXiv:2408.01030.

Presentations

Poster presentation October 2026
S. Majumder and T. Gorda.
Thermal corrections to the equation of state of dense QED at high perturbative orders.
American Physical Society Division of Nuclear Physics Fall Meeting, Philadelphia, Pennsylvania.

Poster presentation August 2026
S. Majumder and T. Gorda.
The N^3LO Mixed Contribution to the Pressure of a Hot and Dense Relativistic QED Plasma.
Department of Physics Poster Session, The Ohio State University, Columbus, Ohio.

Research Experience

The Ohio State University 2026–Present
Graduate Research Associate, research supervisor: Dr. Tyler Gorda.

Tata Institute of Fundamental Research, Hyderabad 2024–2025
Visiting Research Student, research supervisor: Prof. Mustansir Barma.

Yukawa Institute for Theoretical Physics, Kyoto University 2023–2024
Visiting Research Student, research supervisor: Dr. Naoki Sasakura.

Teaching Experience

The Ohio State University — *Graduate Teaching Associate*

Electricity and Magnetism, Waves, Optics, and Modern Physics Spring 2026, Autumn 2025

Indian Institute of Science Education and Research Kolkata — *Teaching Assistant*

Statistical Mechanics Spring 2025, Spring 2024

Probability and Statistics II Spring 2025

Condensed Matter Physics Autumn 2024

Academic Service

President, Physics Graduate Student Council 2025–2026
The Ohio State University.

Leadership team, Science Club 2021–2022
Indian Institute of Science Education and Research Kolkata.

Awards and Honors

Graduate Associate Teaching Award nomination 2025
The Ohio State University.

Silver Medal, University Physics Competition 2022

Finalist, International Astronomy and Astrophysics Competition 2021

References

Dr. Tyler Gorda
Department of Physics and Center for Cosmology and AstroParticle Physics (CCAPP),
The Ohio State University, United States.

Prof. Mustansir Barma
Tata Institute of Fundamental Research, Hyderabad, India.

Dr. Naoki Sasakura
Yukawa Institute for Theoretical Physics, Kyoto University, Japan.

Prof. Arindam Kundagrami
Department of Physical Sciences,
Indian Institute of Science Education and Research Kolkata, India.

Software and Programming

Python, Mathematica, LaTeX.

Languages

Bengali: mother tongue.

Hindi and English: bilingual proficiency.